School Administration and Parent Management System

Functional Specification Document | Version 1.0 | 1 November 2026 | Draft for Review

This document defines functional behavior and access rules for Admin, Staff, and Parent users. It covers teacher and student records, attendance, question-paper approval, parent services, fees, and teacher payroll. Proposed implementation details are marked where relevant.

1. Project Overview

1.1 Purpose

Provide one school management system for administrative, academic, parent-facing, transport, payment, and payroll operations.

1.2 Objectives

- Centralize teacher, student, and attendance records.
- Enforce role-based access with school-wide and assignment-specific scopes.
- Support staff-created, admin-approved question papers.
- Provide parents with bus location, report cards, leave, holidays, fee structures, and payments.
- Let Admin manage teacher salary, payslips, salary status, and loss of pay (LOP).

2. Roles and Permissions

Role	Scope
Admin	School-wide teacher and student management; attendance; question-paper approval; fees; payroll.
Staff	Edit authorized student records, take assigned student attendance, create and submit question papers.
Parent	View only linked children and associated bus, reports, leave, holidays and fees; initiate fee payments.

Capability	Admin	Staff	Parent
Add / edit teachers	Yes	No	No
Add students	Yes	No	No
Edit students	All	Assigned only	No
Student attendance	All	Assigned only	View linked child
Teacher attendance	Yes	No	No
Create question paper	Proposed	Yes	No
Approve question paper	Yes	No	No
Bus / report cards / leave / holidays	Manage*	Scoped*	View linked child
Fee structure	Manage	No	View

Capability	Admin	Staff	Parent
Pay school fees	No*	No	Yes
Teacher salary, LOP, payslips	Yes	No	No

* Proposed capability or ownership to confirm; the core explicitly requested functions are otherwise captured.

3. Functional Requirements

3.1 Authentication and Authorization - FSD-AUTH-001

- FR-AUTH-001: Authenticate Admin, Staff, and Parent accounts.
- FR-AUTH-002: Resolve assigned roles and show permitted navigation, screens, and actions.
- FR-AUTH-003: Enforce role and record-level authorization on backend APIs, independent of frontend controls.
- FR-AUTH-004: Restrict parents to linked children and staff to assigned students.
- FR-AUTH-005: Reject unauthorized requests and audit security-sensitive operations.

3.2 Teacher Management - FSD-TCH-001

- FR-TCH-001: Admin creates teacher records.
- FR-TCH-002: Admin edits and views teacher profiles.
- FR-TCH-003: Staff and Parent cannot edit teacher employment details.

Suggested field	Purpose
Teacher ID / employee code	Unique identity
Name and contact	Profile and communication
Department / subject / assigned class	Academic assignment
Joining date / employment status	Employment administration
Salary information	Restricted payroll data

3.3 Student Management - FSD-STD-001

- FR-STD-001: Admin creates and edits student records school-wide.
- FR-STD-002: Staff edits permitted fields for assigned students only; Staff cannot add students.
- FR-STD-003: Parent cannot modify official student records.
- FR-STD-004: Each student may be associated with authorized parent accounts.

Suggested fields: admission number, name, birth date, class, section, roll number, academic year, parent association, enrollment status.

3.4 Student Attendance - FSD-ATT-001

- FR-ATT-001: Admin marks, views, and edits attendance for any student.
- FR-ATT-002: Staff marks and edits attendance for assigned students only.
- FR-ATT-003: Parent views linked child attendance without edit access.
- FR-ATT-004: Prevent duplicate records for student, date, and session.
- FR-ATT-005: Audit changes with actor, timestamp, and prior value.

Suggested attendance statuses: Present, Absent, Late, Leave.

3.5 Teacher Attendance - FSD-ATT-002

- FR-ATT-006: Admin views, marks, and edits teacher attendance.
- FR-ATT-007: Teacher attendance feeds authorized payroll and LOP calculations.
- FR-ATT-008: Staff cannot modify their official teacher attendance record.

4. Question Paper Management - FSD-EXAM-001

Staff creates question papers; Admin reviews submissions and approves or rejects them.

- FR-EXAM-001: Staff creates question papers with subject, class, examination, and academic year.
- FR-EXAM-002: Staff saves drafts and submits papers for approval.
- FR-EXAM-003: Admin reviews pending papers and records an approve or reject decision.
- FR-EXAM-004: Rejected papers return to Staff for revisions and resubmission.
- FR-EXAM-005: Maintain decision history and keep unapproved papers from approved release.
- FR-EXAM-006: Staff cannot approve their own papers.

State	Allowed next actions
Draft	Staff edits or submits
Pending Approval	Admin approves or rejects
Rejected	Staff edits and resubmits
Approved	Approved material is available for authorized release

Workflow: Staff creates paper -> saves draft -> submits -> Admin reviews -> approves, or rejects with comments -> Staff revises -> resubmits.

5. Parent Portal - FSD-PAR-001

5.1 Dashboard

Show linked student and class, attendance summary, report-card availability, bus tracking, upcoming holidays, leave information, unpaid fees, and payment status.

5.2 Live Bus Tracking - FSD-BUS-001

- FR-BUS-001: Parent sees the assigned school bus on a map when valid GPS data is available.
- FR-BUS-002: Display most recent update time and stale or unavailable state.
- FR-BUS-003: Restrict location access to buses or routes associated with linked children.

External dependency: a GPS device or authorized location provider is required. Provider and refresh frequency are not yet selected.

5.3 Report Cards - FSD-RPT-001

- FR-RPT-001: Parent views only published reports for linked children.
- FR-RPT-002: Show examination, academic period, subjects, grades or marks, and student identity.
- FR-RPT-003: Hide draft and unpublished reports.

5.4 Leave Information - FSD-LVE-001

- FR-LVE-001: Parent views leave records, dates, available reasons, and status for linked children.
- FR-LVE-002: Parent cannot modify official attendance or leave through this read-only view.

5.5 School Holidays - FSD-HOL-001

- FR-HOL-001: Parent views published holidays with date, title, and academic year.
- FR-HOL-002: Present the calendar chronologically and hide unpublished entries.

6. School Fees Management - FSD-FEE-001

6.1 Fee Structure

- FR-FEE-001: Admin creates, updates, and assigns applicable fee structures.
- FR-FEE-002: Parent views linked child fee components, due amounts, and dates.
- FR-FEE-003: Audit fee-structure changes.

Proposed categories: tuition, examinations, transportation, miscellaneous.

6.2 Fee Payment

- FR-FEE-004: Parent views pending fees and initiates eligible payments.
- FR-FEE-005: Backend creates a transaction and independently verifies gateway success.
- FR-FEE-006: Verified payments update the outstanding balance and generate a receipt.
- FR-FEE-007: Show pending/failed states correctly and protect against duplicate settlement.
- FR-FEE-008: Parent views payment history and receipts.

Workflow: Parent views fee -> initiates payment -> provider processes -> backend verifies -> success updates ledger and receipt; failed/pending outcomes remain unsettled.

7. Teacher Payroll Management - FSD-PAY-001

7.1 Salary Structure

- FR-PAY-001: Admin maintains teacher salary components and monthly payroll data.
- FR-PAY-002: Admin calculates and reviews payroll under school-approved policies.
- FR-PAY-003: Record salary changes; restrict payroll data to Admin.

Payroll field	Meaning
Teacher / payroll month	Employee and period
Basic salary / allowances	Earnings
Deductions / LOP days / LOP amount	Reductions
Net salary / payment status	Final amount and settlement state

7.2 Loss of Pay - FSD-LOP-001

- FR-LOP-001: Admin reviews teacher attendance for payroll month.
- FR-LOP-002: Admin records or adjusts LOP days with audit history.
- FR-LOP-003: Calculate deduction based on approved eligible salary, payroll divisor, and LOP days.
- FR-LOP-004: Require review before finalizing and authorized correction afterward.

Illustrative formula: LOP deduction = (LOP-eligible monthly salary / applicable payroll days) x LOP days. Exact policy, leave rules, and rounding require school approval.

7.3 Payslips - FSD-PSL-001

- FR-PSL-001: Admin generates payslips from finalized payroll.
- FR-PSL-002: Payslip lists teacher details, month, earnings, deductions, LOP, and net salary.
- FR-PSL-003: Admin views/downloads payslips; deny unauthorized access.

7.4 Salary Payment Status - FSD-SAL-001

Status	Meaning
Draft	Calculation in progress
Calculated	Computation complete
Approved	Authorized for payment
Payment Pending	Settlement not confirmed
Paid	Payment confirmed
Payment Failed	Attempt unsuccessful

External bank disbursement integration is not assumed.

8. Frontend and Backend Architecture

Module	Frontend	Backend
Auth	Login, navigation, route guards	Authentication, trusted authorization
Teacher / Student	Forms, lists, scoped UI	CRUD, assignment checks
Attendance	Date and student attendance grid	Validation, records, audit
Question papers	Editor, submission, review	State transitions, decision authorization
Bus and reports	Map and published results	GPS data, child access filtering
Fees	Fee display, checkout, receipts	Ledger, verification, reconciliation
Payroll	Salary, LOP, payslip forms	Salary rules and private records

8.1 Proposed API Interfaces

Method	Endpoint	Access
POST	/api/teachers	Admin
PATCH	/api/teachers/:id	Admin
POST	/api/students	Admin
PATCH	/api/students/:id	Admin / scoped Staff
POST	/api/students/attendance	Admin / scoped Staff
POST	/api/teachers/attendance	Admin
POST	/api/question-papers	Staff; Admin proposed
POST	/api/question-papers/:id/submit	Paper owner
POST	/api/question-papers/:id/decision	Admin
GET	/api/parents/children/:id/bus	Linked Parent
GET	/api/parents/children/:id/reports	Linked Parent
GET	/api/parents/children/:id/fees	Linked Parent
POST	/api/fee-payments	Linked Parent
POST	/api/payroll/calculate	Admin
PATCH	/api/payroll/:id/lop	Admin
GET	/api/payroll/:id/payslip	Admin

These endpoints are candidate contracts, not pre-existing APIs. Every handler must validate identity, role, and record ownership/assignment using trusted server-side information.

9. Data Entities

Entity group	Entities and relationships
Identity and academics	User, Role, Permission, Teacher, Student, ParentStudent, Class, Section
Attendance and exams	StudentAttendance, TeacherAttendance, QuestionPaper, QuestionPaperApproval, ReportCard
Transport and calendar	SchoolBus, Route, BusLocation, Holiday, StudentLeave
Finance and audit	FeeStructure, StudentFee, FeePayment, SalaryStructure, Payroll, PayrollAdjustment, Payslip, AuditLog

10. Nonfunctional Requirements

- NFR-001: Responsive UI for desktop, tablet, and mobile.
- NFR-002: Authentication, backend authorization, and sensitive-data protection in transit and at rest.

- NFR-003: Audit attendance edits, exam decisions, payment adjustments, and payroll changes.
- NFR-004: Consistent dates, currency, academic-year context, and actionable error states.
- NFR-005: Payment idempotency, retry handling, and reconciliation.
- NFR-006: Bus tracking displays location freshness.
- NFR-007: Define backup, recovery, retention, and performance targets before release.

11. Acceptance Criteria

ID	Scenario	Expected
TC-001	Admin adds teacher	Created
TC-002	Staff attempts teacher creation	Denied
TC-003	Admin edits any student	Permitted
TC-004	Staff edits assigned student	Permitted
TC-005	Staff edits unassigned student	Denied
TC-006	Staff takes assigned attendance	Saved
TC-007	Staff edits teacher attendance	Denied
TC-008	Staff submits question paper	Pending Approval
TC-009	Admin approves paper	Approved
TC-010	Admin rejects paper	Returned for revision
TC-011	Staff approves own paper	Denied
TC-012	Parent views linked child report	Published report visible
TC-013	Parent accesses unrelated report	Denied
TC-014	Bus data stale	Warning visible
TC-015	Verified payment succeeds	Ledger and receipt updated
TC-016	Gateway remains pending	Not marked paid
TC-017	Admin applies LOP	Approved deduction reflected
TC-018	Parent accesses teacher payslip	Denied

Document status: Draft for stakeholder review. Confirm proposed ownership, integrations, and payroll policies before treating this as a final implementation contract.